INCH-POUND
MIL-C-83513/20A
14 Oct 1993
SUPERSEDING
MIL-C-83513/20
27 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, SOCKET CONTACTS, 3 ROW ,SOLDER TYPE, STANDARD PROFILE, 51 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification Listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.

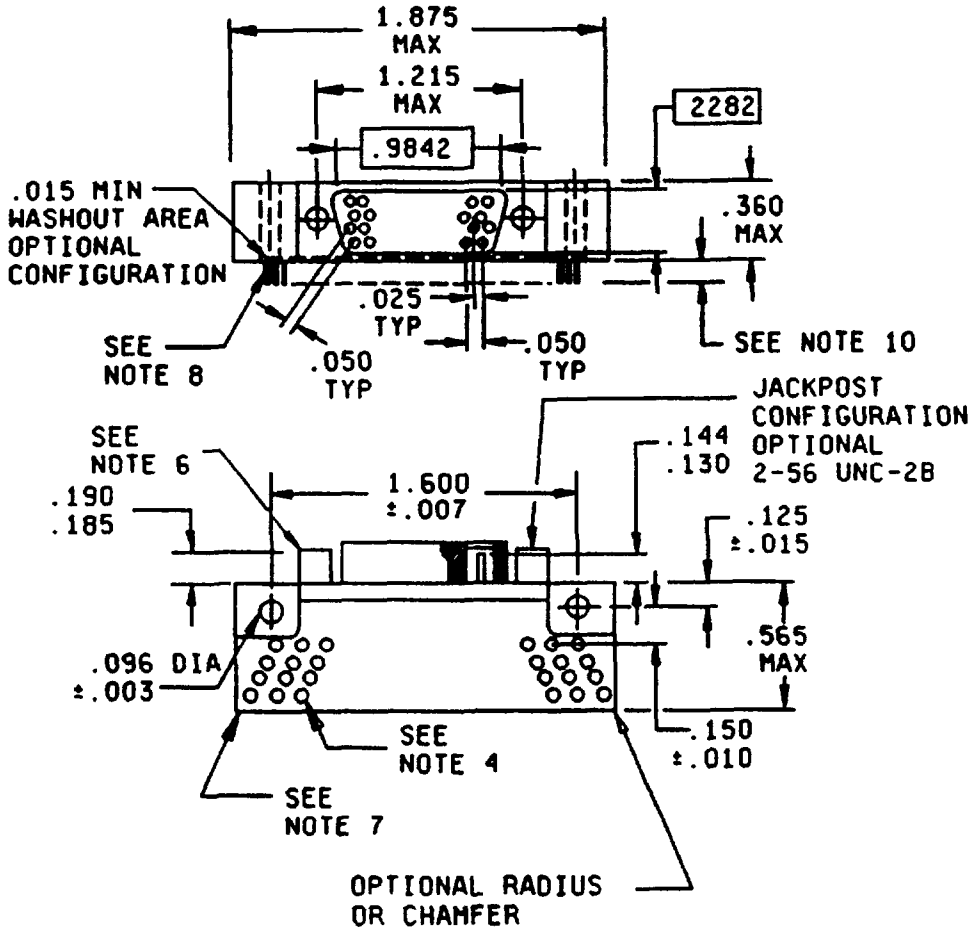
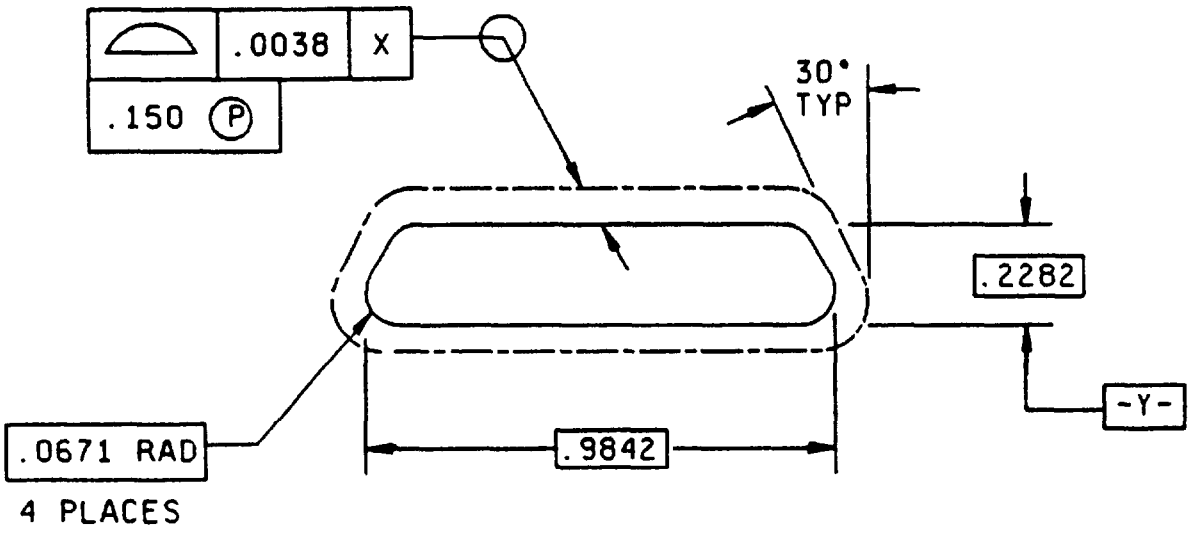


FIGURE 1 Connector, receptacle, .050 spacing

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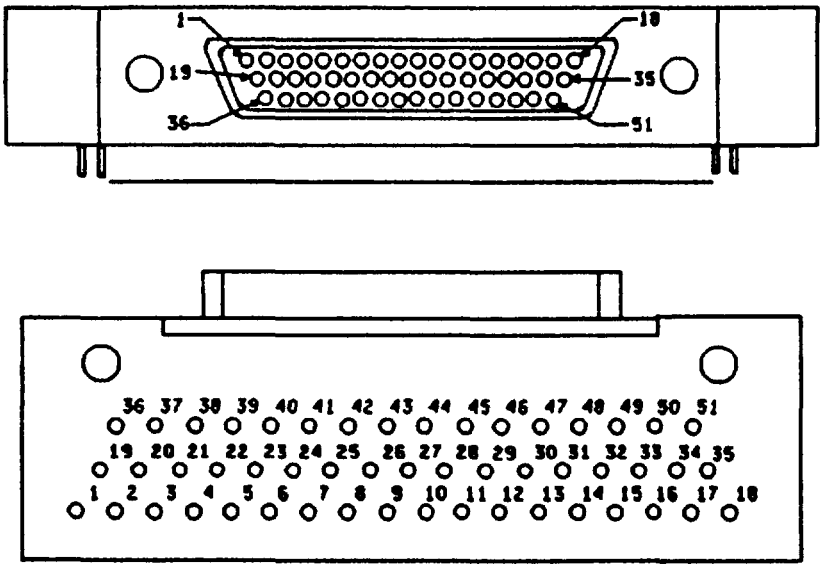
DETAIL OF INTERFACE

Inches	mm	Inches	mm	Inches	mm
.003	0.08	0671	1.704	.360	9.14
.0038	0.097	.096	2 44	.565	14.35
.007	0 19	.125	3.18	.9838	24.999
.010	0.25	.150	3.81	1.215	30.86
.015	0.38	185	4 70	1.600	40.64
.025	0.64	.190	4.83	1 875	47.63
.050	1.27	.2282	5.796		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm)
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513 Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts.
6 Jackpost (permanently attached), when specified Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35
7. Separately molded plastic body (if used) shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519
8 Socket termination wires shall conform to QQ-W-343, no. 24 AWG copper (solid wire)
9 Interfacial seals shall included with each type receptacle connector
10 Termination lengths available 109 (2 77 mm), 140 (3 56 mm), or 172 (4 37 mm) The tolerance shall be ± 015 (0.38 mm) for all termination lengths

FIGURE 1 Connector, receptacle, 050 spacing - Continued.

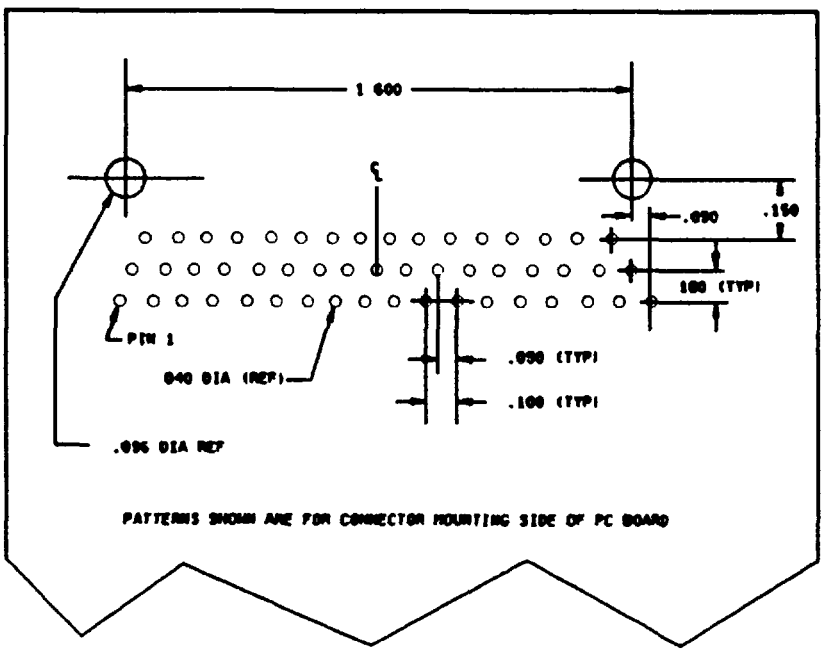
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TERMINATION VIEW-RIGHT ANGLE
STANDARD PROFILE

NOTE: Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2. Insert arrangement.



Inches	mm
.040	1.02
050	1.27
.096	2.44
.100	2.54
.150	3.80
1.600	40 64

FIGURE 3 Layout arrangement.

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REQUIREMENTS

Dimensions and configurations See figures 1 and 2.

Current rating, maximum 3 amperes per contact.

Materials

Termination organization area. Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513

Shell The requirements for shell materials shall be in accordance with MIL-C-83513

Plastic body or plastic molding Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519

Jackpost Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.

Wire termination pins Wire termination pins shall conform to QQ-W-343, no 24 AWG copper

Mating connector· Shall conform to MIL-C-83513/1 and MIL-C-83513/3

Printed circuit board layout: See figure 3.

Plating of termination leads Solder dipping socket of termination leads will be accomplished in Sn60 Pb40 or Sn63 Pb37 in accordance with QQ-S-571.

Part or Identifying Number (PIN): PIN shall consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code for the shell finish and the hardware.

M83513/20-	G	01	C	P
Specification sheet number	Insert arrangements (see figure 2) G = 51	Termination length 01 = .109 02 = 140 03 = .172	Shell finish C = cadmium N = electroless nickel (space application)	Hardware N = no jackpost P = jackpost attached

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - AS, SH
Air Force - 17, 99
NASA - NA

User activities
Army - AT, ME
Navy - CG, MC

Preparing activity
Air Force - 85

Agent·
DLA - ES

(Project 5935-3921-11)

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